

SCITRAM: Single-cell Integrated transcription regulation modeler

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Tutorial

Introduction

Accessing to the molecular cell heterogeneity thanks to the development of single-cell omics technologies is revolutionizing our understanding of complex living systems. While generating such type of data is becoming a streamline process, data processing remains to be possible either by command-line driven tools, or commercial solutions. Furthermore, single-cell transcriptomics data processing is usually reduced to cells stratification on the grounds of gene expression signatures, while aspects like deconvolving the various gene regulatory programs responsible for cells heterogeneity are not classically addressed with the available tools, or performed under different data processing platforms.

Herein we introduce SCITRAM, a computational solution able to (i) process multiple single-cell transcriptomics files at once, (ii) perform dimensionality reduction while retaining the origin of cells retrieved on the obtained clusters, (iii) compute differential gene expression profiling among stratified clusters; (iv) annotate clusters based on the use of user-provided cell-type gene association information; (v) reconstitute gene regulatory networks per cluster by using transcription factors-target gene relationships; and (vi) classify transcription factors as master drivers defining cluster differences retrieved within single-cell transcriptomics readouts.

All the aforementioned steps are possible within an intuitive user-friendly environment, making of SCITRAM a unique platform providing a large number of data processing options. Furthermore, SCITRAM distribution is meant to avoid tedious tools/dependencies installations, relying in a direct executable distribution format for both windows and Mac, in addition to the regular Linux distribution.

The screen shots showed here corresponds to a real analysis using the data mentioned in the main publication and provided by the team of [Sonia Guedan](#) (Institut d'Investigacions Biomèdiques Agustí Pi i Sunyer, Spain) within the frame of the European collaborative project [T-FITNESS](#).

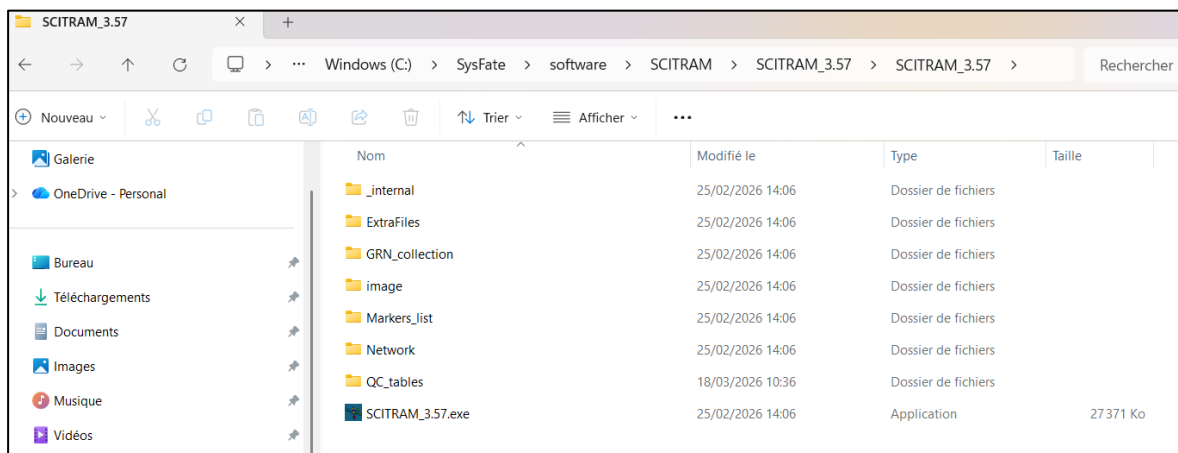
SCITRAM installation

Windows / OSX Installation

We provide 3 versions, one for Microsoft windows (working under versions 10 and 11), one for OSX (Mac) and one python script for Linux and other systems.

To use SCITRAM under **Windows** system you have to:

- 1) Download the SCITRAM_vxx.zip folder following the next link:
<https://github.com/SysFate/SCITRAM>
- 2) De-compress the file.
- 3) Double-click on the SCITRAM_vxx.exe executable file.



Is important to keep the directories in the same place that the executable file (.exe). We include some files inside directories and will be explained in the corresponding sections.

SCITRAM under OSX:

In preparation

Linux installation

1. Go to <https://www.anaconda.com/download>
2. Register your account or simply skip it.
3. Chose the Linux distribution and select the 64-bit (x86) installer



Alternatively, you can go directly to download the file in your shell window, type:

```
wget https://repo.anaconda.com/archive/Anaconda3-2024.10-1-Linux-x86_64.sh
```

4. Open your shell window, browse to “Downloads” and type :

```
sh Anaconda3-2024.10-1-Linux-x86_64.sh
```

Follow the instructions.

*Conda versions can change, the file name could resemble to Anaconda3-20XX.XX-X-Linux-x86_64.sh

5. Go to our github: <https://github.com/SysFate/SCITRAM> and download the files SCITRAM_vxx.py and the file SCITRAM_requirements.yml
6. Create and activate new environment typing

```
conda create -n SCITRAM -f SCITRAM_requirements.yml  
conda activate SCITRAM
```

7. Go to the SCITRAM directory and run the python script:

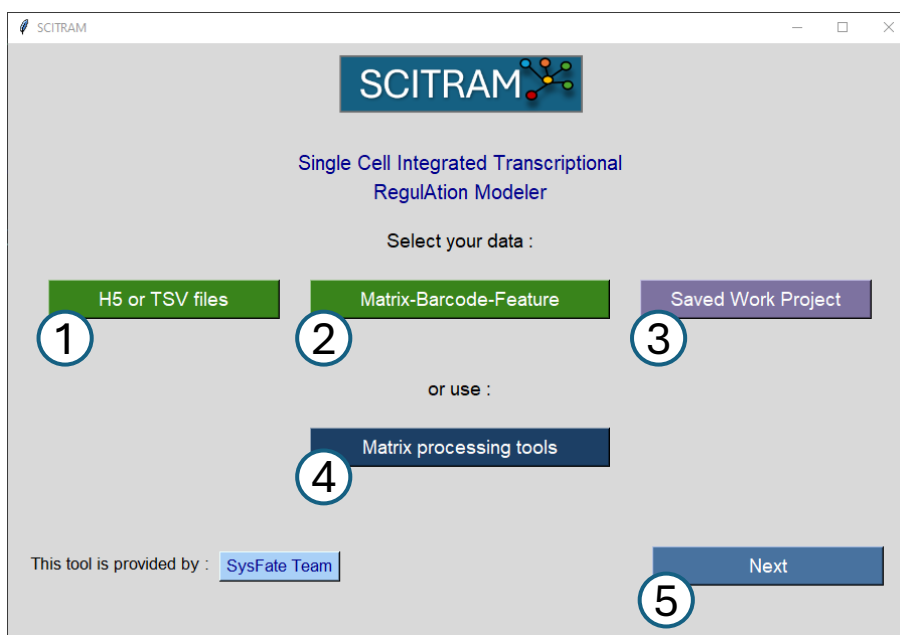
```
python SCITRAM_vxx.py
```

Reading Single Cell Files

This tool was developed to read the most common files: hierarchical data format (.H5), matrix in plain text delimited by tabs (.tsv), and files in format [matrix.mtx - barcodes.tsv, -features.tsv], You can upload one or more files.

Your datasets need to be preprocessed by tools generally provided by the company commercializing the reagents used for the single-cell assays. For example, 10 X genomics provide the tool Cell ranger, which will generate H5 or matrix-barcodes-features files using the genome pre-build for this platform. Some sequencing platforms already provide this files, but if you have only the “fastq” files you can search how to do this pre-process for example in [Cloud Analysis | 10x Genomics](#). or [Installing Cell Ranger - 10x Genomics](#). Repositories like NCBI GEO provide files in SCITRAM compatible formats.

Upload files window (1)



Select your H5 or plain text in **(1)**, you can select many files as you want (that depends on the capacity of your computer). If you have one table with counts for the plain text file, the gene names need to be placed in the first column, and the “name of cells” (barcodes) need to be in the first row. Gene names should be the “gene symbol” approved by the HGNC, this is for example actin beta: “ACTB” and Not: “ENSG00000075624”. For the mouse genome the accepted format should be “Actb”. Text separators could be tab (.tsv) or comma (.csv) delimited.

In **(2)** you can upload the data with the system of 3 files, note that if you want to select one directory with many samples each sample needs to have the same name, only the extension changes (ex. sample1_matrix.mtx, sample1_barcodes.tsv, and sample1_features.tsv)

You can continue your previously saved work in **(3)**. We provide a Tool to help in the pre-process of your data **(4)**, explained below.

To continue setup the analysis, click in next **(5)**.

Matrix processing tools supp window

SCITRAM Tools
— □ ×

Matrix File Converter 1

Source Matrix

Output File

Run

Subset Cells 2

Source Matrix

Barcodes

Output File

Run

Select one column

▼

Fuse Multiple Matrix 3

Source Matrix

Output File

Run

Output Format:

H5
▼

We provide a group of tools to help in your analysis before to start the main process. For these 3 tools you need to select your sample, select the output file name and directory, and simply run, in the 3 cases the input and output could have the format “.h5” or “.tsv”. In **(1)** you can interconvert files between 2 types of formats described in the section above. “.H5” <=> “.tsv, .tsv, .txt” . You can subset cells in **(2)** using your matrix and a tab delimited file with the barcodes (cell names), for this you need to select the column that contain the barcodes. You can save time in the main process if you fuse your multiple files in **(3)**.

Setup parameters window

Quality Parameters :

Number of cells per gene: > 3

Number of reads per cell: > 500 < 15000

Number of genes per cell: > 200 < 25000

Filter cells (% mito counts): > 5

☒ Delete doublets Take subsample of data (%): 100

☒ Delete BlackList genes
(Mit, ImmGlob, MHC, Ribo, HemoG)

Normalization :

☒ Log-normalize

☐ Scale Data

☒ Regress out cell-cycle phase

PCA & UMAP :

Number of Highly Variable Genes : 2000

Number of PCs : 20

Number of neighbors : 15

Resolution for leiden clustering : 0.5

Back Start

With the aim of addressing expectations of naïve but also advanced users, SCITRAM proposes a by-default parameter setup, but it gives the flexibility to the user to modify them if wished.

In panel **(1)** you can use different parameters to filter your samples in terms of cell, genes or counts, even more complex filters.

- Number of cells per gene: avoid to have genes present in a few number of cells.
- Number of reads per cell: filter out cells with few counts, but also highest amounts of counts notably because could be consequence of doublets, triplets in the same encapsulated droplet.
- Filter cells (% mito counts): Previously was calculated that one cell have around 5 % of counts from mitochondrial genes, you can filter out cells in mitosis for example or other states where you have more reactive mitochondria. Take in account that some experiments like Multiomes require to separate the nucleus in this case for multiome RNA + ATAC is advised to use the filter 10%.
- Delete doublets: This process take in account the expected number of counts respect to the number of cells and the saturation of the system (data provided by 10x), take in mind that this process take more time because basically run the first process two times.
- Delete Blacklist genes: You can get out of the analysis genes that are irrelevant for your study and/or have very high levels and variables in the cells, for example the immunoglobulins, hemoglobins, ribosomal genes). You can read exactly which genes are deleted in the file Human_blacklist_genes_Ensembl115.tsv 8 or Mouse...) inside the “ExtraFiles” directory. This file could be modified.

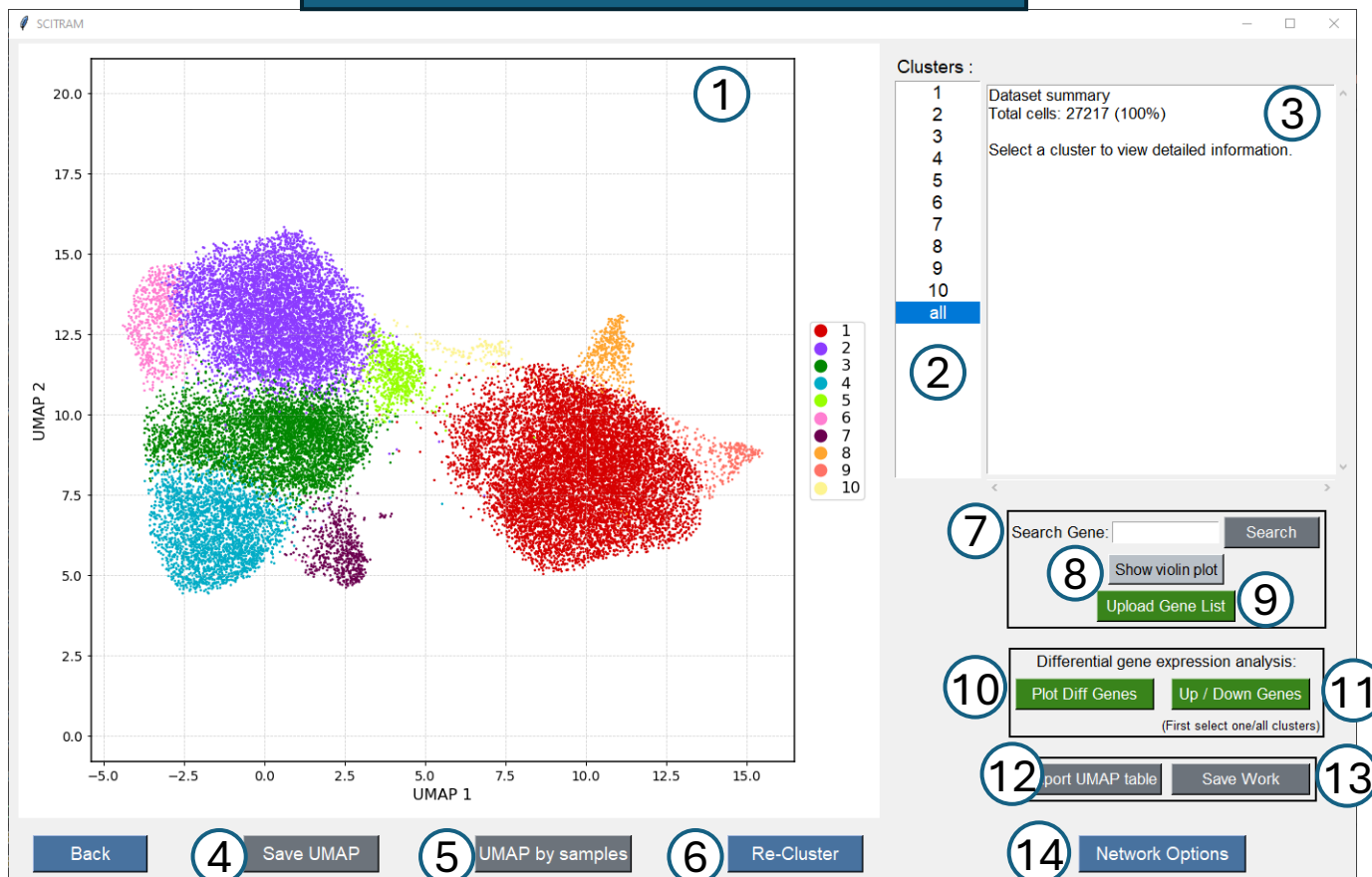
- In panel (2) you can setup the normalization through the cells:
- Log-Normalize: this is a normalization used by default in different packages like Scanpy and Seurat, uses the modified logarithmic normalization to avoid outliers (intra samples) and the Harmony method for different samples (intra samples) this method provide values in log2 scale..
- Scale data: this method will provide a relative number of counts for each gene through the cells, that's is zero for the media of the gene and the difference of the values in log 2 scale.
- Regress out cell-cycle phase: The normalization takes in account the expression of some genes like the cyclins. This can help in experiments where you assume that your cells have the same levels of cell division. If is not the case and you are searching this differences, is better to avoid this process.

Finally, for the PCA and UMAP setup **(3)**, you can modify the way of how information you should use to separate the cells, how intensive should be the analysis and how to group the cells.

- Number of highly variable genes: When you have many cells, the capacity to analyze all the genes to classify these cells could take several hours or crash the computer, even could generate errors, we assume that the most variables genes are around 2,000, maybe if you need to be mor exhaustive you can increase.
- Number of PCAs: you can increase if you want to be more exhaustive in your analysis, but certainly when you increase the number of PCs there are no big differences in the number of clusters obtained.
- Number of neighbors: To group your cells you can assume that your cells are very connected between them (high values) or less connected (lower values).
- Resolution of Leiden clustering: maybe the most important of these parameters, because you can increase or decrease the number of clusters, however because this is a non supervised analysis, you can't generate a specified number, to obtain more clusters you can increase the resolution. This parameter generally you can change it after one first analysis if you are sure that you have one or mor subclusters inside one. We provide one tool in the main window to help you in this work.

Main working window. Here you have many ways to visualize your sample(s) in graphics and tables. At the end you can decide which clusters will be compared to obtain the gene regulatory network.

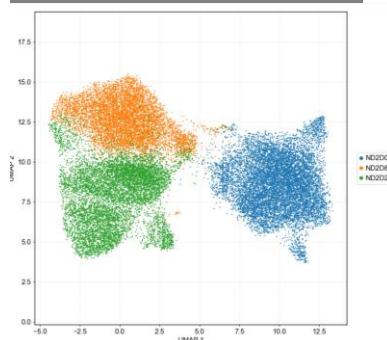
Main window



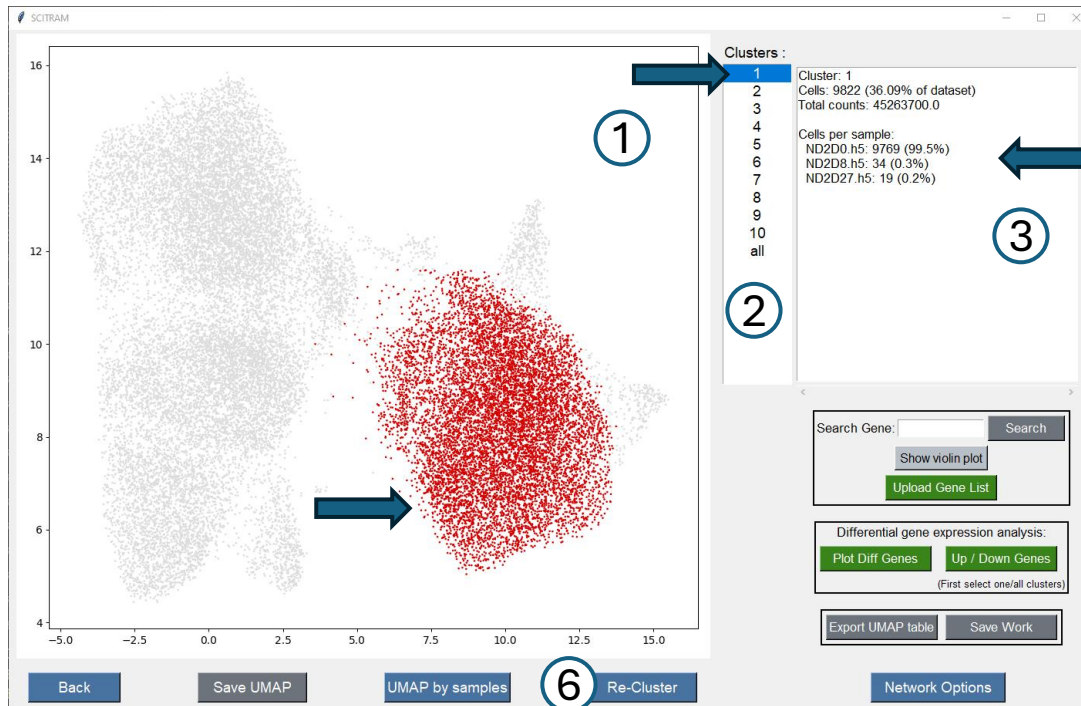
In (1) you can visualize the UMAP in different ways, modified by other analysis. All the time you can save the graphic showed here with the button Save UMAP (4).

SCITRAM allows to visualize cells stratification based on the dataset of origin (5):

5 UMAP by samples



By selecting a given cluster **(2)**, the user can get the number of cells per cluster as well as its stratification based on the different uploaded data **(3)**:



In cases in which a further cluster stratification is need, SCITRAM provides a re-clustering option **(6)**

Re-cluster supp window

6 Re-Cluster

Local Re-Clustering

Select one cluster:

1

PCA & UMAP:

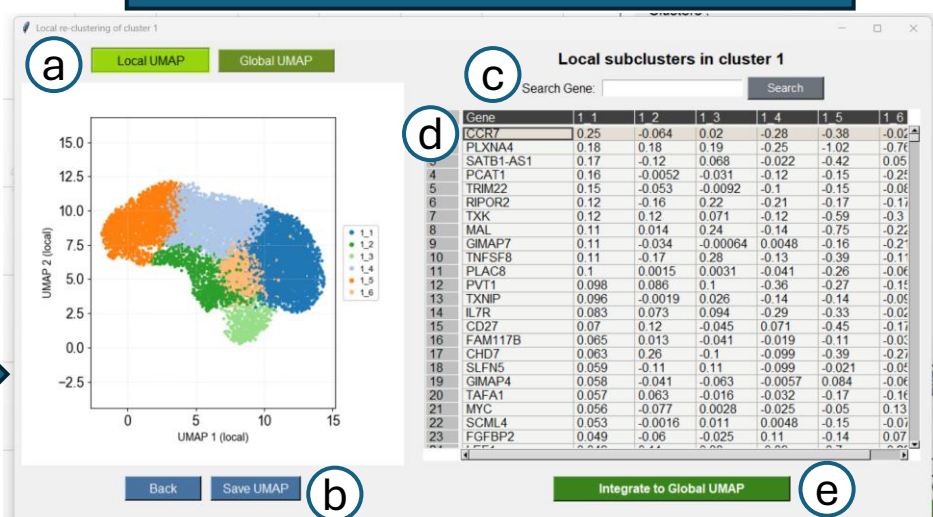
Number of Highly Variable Genes: 1500

Number of PCs: 15

Number of neighbors: 10

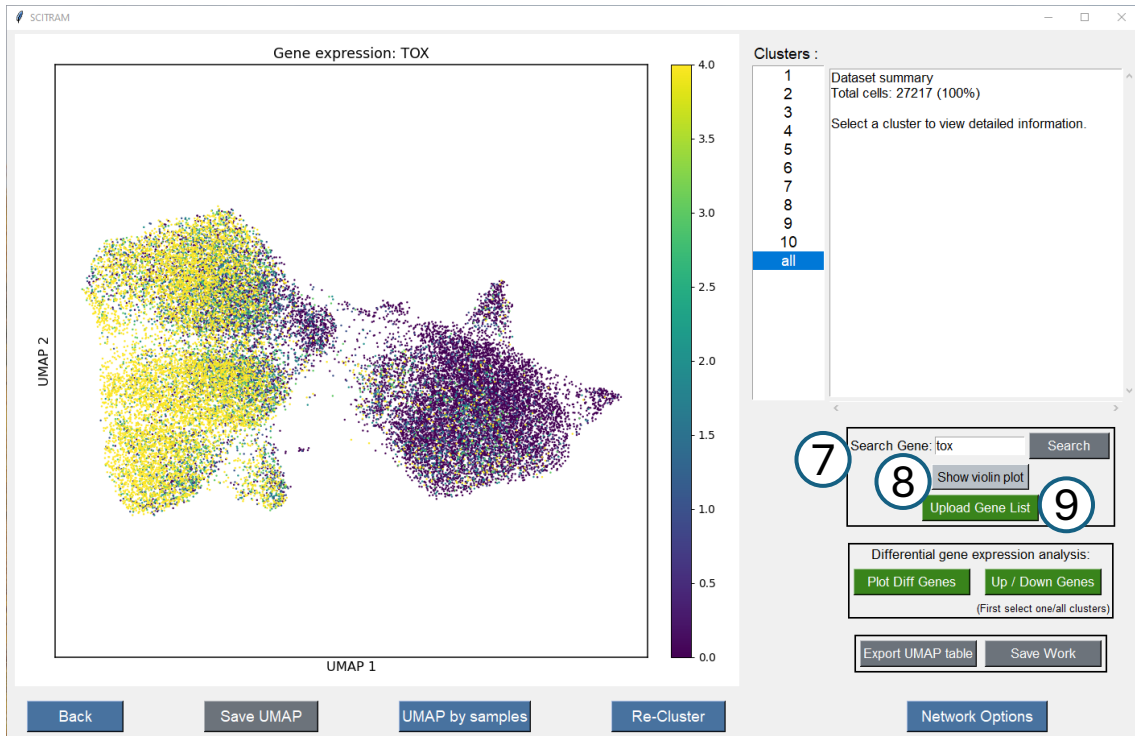
Resolution for leiden clustering: 0.4

Re-Clustering

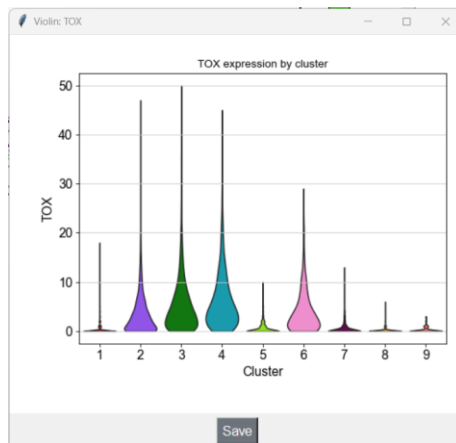


You can see how the cells are clustered in a local or global way (upper buttons) **(a)** and you can see how is expressed certain gene in **(c)**. The table below **(d)** is interactive, and you copy paste to an excel file. The re-clustered map can be saved as image **(b)**, also be integrated within the global UMAP for proceeding with the analysis **(e)**, or it could be saved as a separate object.

SCITRAM allows to visualize the expression levels of a given gene by using a query panel **(7)**.



Gene expression levels can also be displayed in a violin plot format **(8)**:



Violin plot supp window

Furthermore, when larger list of genes are wished to be analyzed users can upload a Gene list in **(9)** where the first column corresponds to a “cell marker type” annotation (or any other annotation of interest) and the second to the genes ID of interest (official gene symbols) see example below. Important: the columns needs to be named: 'marker_type' and 'gene'.

Once uploaded, the list of genes and its corresponding annotation is displayed in a table where the frequency of expression of each gene per cluster is displayed and could be saved as excel format **(c)**:

9 Query list supp window

7 Search Gene: Search

8 Show violin plot

9 Upload Gene List

10 Differential gene expression analysis:
Plot Diff Genes Up / Down Genes
(First select one/all clusters)

12 port UMAP table Save Work

14 Network Options

SCITRAM

All Clusters

	marker_type	gene	cluster 1	cluster 2	cluster 3	cluster 4	cluster 5	cluster 6	cluster 7	cluster 8	cluster 9
6	Effector_markers	PRF1	0.87	2.57	1.52	2.01	1.37	1.12	1.45	0.47	0.59
7	Effector_markers	GZMB	23.27	29.32	10.20	12.53	18.06	5.85	59.67	13.11	8.73
8	Effector_markers	GNLY	2.71	24.14	40.20	43.79	36.75	38.64	2.43	1.41	7.75
9	Exhaustion_markers	PDCD1	0.0069	0.33	0.19	0.084	0.091	0.16	0.064	0.0074	0.038
10	Exhaustion_markers	TOX	0.36	3.65	5.36	6.32	0.57	3.96	0.46	0.14	0.23
11	Exhaustion_markers	CXCL13	0.0012	0.42	1.99	0.52	0.068	0.79	0	0.0037	0.06
12	Exhaustion_markers	TIGIT	0.12	1.98	2.53	2.29	0.96	1.72	0.1	0.22	0.31
13	Exhaustion_markers	CTLA4	0.026	0.94	1.29	1.89	0.38	0.83	0.038	0.015	0.11
14	Exhaustion_markers	TNFRSF9	0.018	2.81	1.08	1.18	0.19	0.98	0.63	0.0074	0.34
15	Exhaustion_markers	HAVCR2	0.57	1.66	1.91	0.67	1.36	1.41	1.45	0.53	0.2
16	Exhaustion_markers	LAG3	0.27	2.16	1.25	1.10	1.09	0.8	0.64	0.35	0.44
17	Exhaustion_markers	NR4A1	0.016	0.51	0.34	0.27	0.29	0.59	0.16	0.0074	0.09
18	Tex_GZMKpos	CD28	0.1	0.16	0.032	0.28	0.05	0.08	0.12	0.089	0.015
19	Tex_GZMKpos	EOMES	0.0084	0.01	0.0064	0.0011	0.016	0.0051	0.026	0.0074	0.007
20	Tex_GZMKpos	CCR5	0.099	0.17	0.12	0.11	0.24	0.09	0.047	0.052	0.038
21	Tex_GZMKpos	CCL4	0.81	9.69	6.45	2.87	11.36	4.26	27.74	0.54	2.81
22	Tex_GZMKpos	CCL5	6.12	5.98	12.31	5.96	9.48	11.16	5.80	10.11	2.53
23	Tex_GZMKpos	CD200R1	0.058	0.018	0.14	0.026	0.11	0.16	0.031	0.019	0.015

(No Title)

Show on UMAP

Include Heatmap

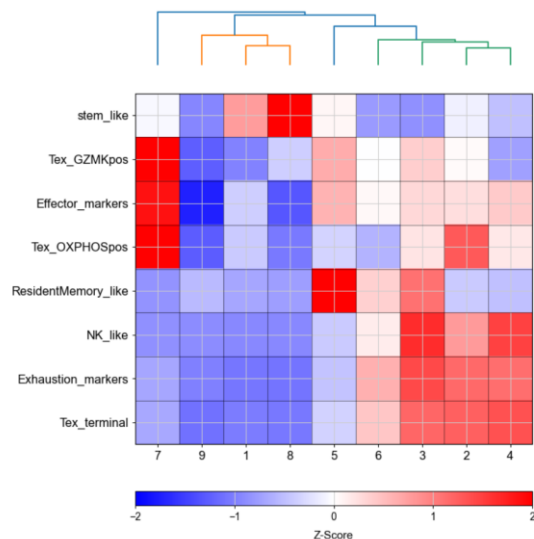
Export table

Heatmap all markers

All markers heatmap supp window

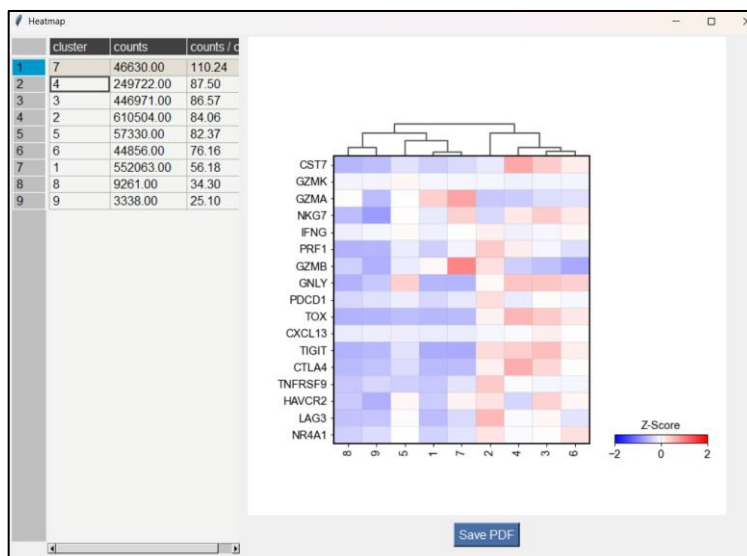
By clicking on **(d)** SCITRAM provides a global view of the enrichment of the queried cell markers across the different clusters.

In this case, you can see that exhaustion markers are preferentially enriched on clusters 6, 3, 2 and 4; effector markers in cluster 7 and stem-like markers in cluster: 8.

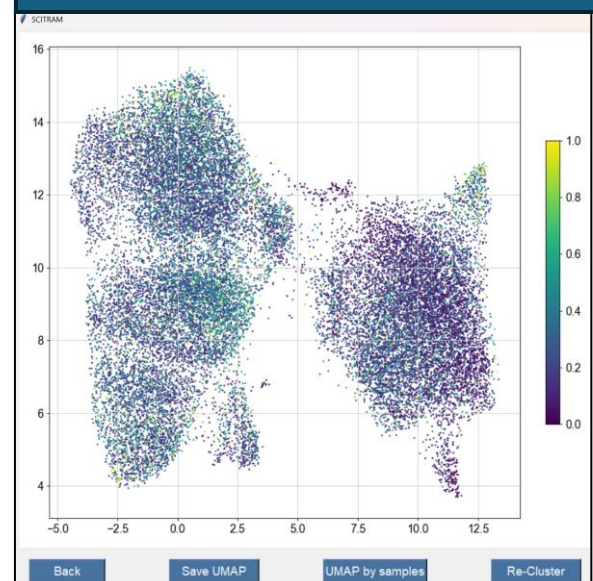


We can also select some rows on the table to obtain a view of the selected genes per cluster heatmap and their display within the main UMAP **(a)** and **(b)**

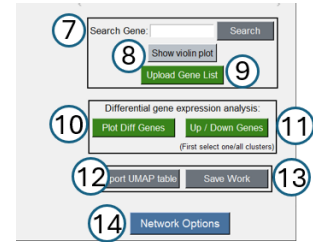
Selected genes heatmap supp window



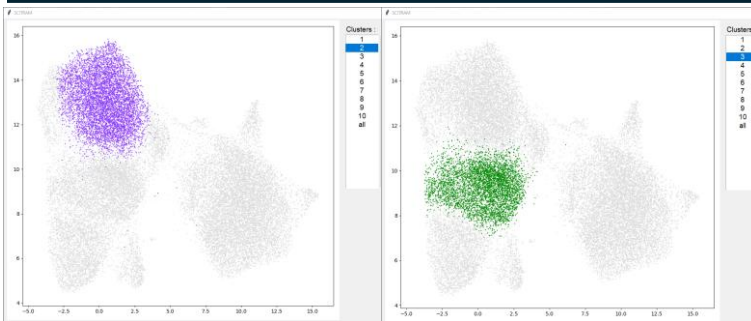
Main window



To perform a differential expression analysis between clusters, the user is expected to select the clusters to compare based on their own criteria. In this case, we will compare cluster 3 (97% cells issued from day 27 in CAR-T cells treatment;) vs cluster 2 (96% cells from day 8).



Main window: selected clusters 3 and 2



10 Plot Diff Genes supp window 1/2

Reference cluster(s):

1

2

3

4

5

6

7

8

9

10

Target cluster(s):

1

2

3

4

5

6

7

8

9

10

LFC > 0.5

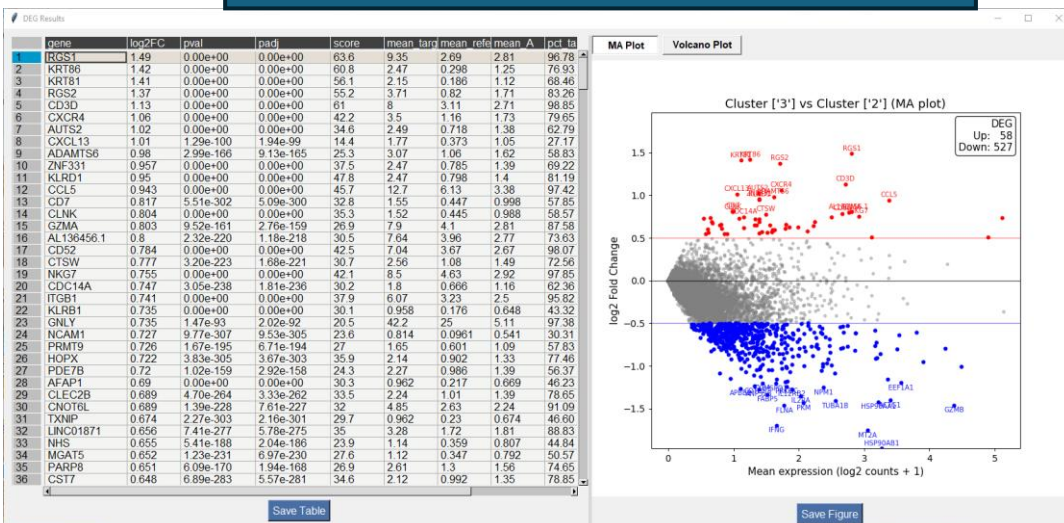
p-value < 0.05

Top N genes: 20

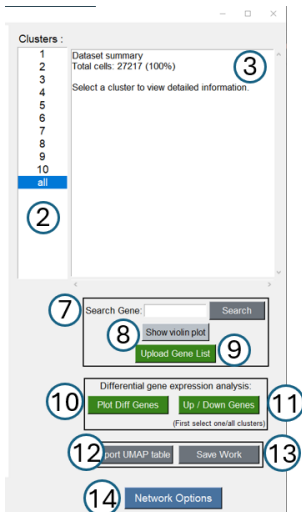
Analyze

A new window will be open after the analysis in the plot diff genes supp window 1/2 (sowed below). In this window you can obtain the table and save in tsv or excel formats, at right you can see the MA plot, you can choose the volcano plot also in the top level of the graphic. The colored points depends on the LFC and P-Value (U test) setup in the window 1, notably you can position your cursor over the points of the graphic and you can see the name of the gene. The number of gene labels depends on the parameter “Top N genes”.

Plot Diff Genes supp window 2/2



SCITRAM can also provide a direct view of all UP/Down regulated genes **(11)** across all clusters at once if « all » is selected; or for a given selected cluster in **(2)**:



Clusters:

Dataset summary
Total cells: 27217 (100%)

Select a cluster to view detailed information.

1
2
3
4
5
6
7
8
9
10
all

7 Search Gene: Search

8 Show violin plot

9 Upload Gene List

10 Differential gene expression analysis:
Plot Diff Genes Up / Down Genes
(First select one/all clusters)

12 Sort UMAP table

13 Save Work

14 Network Options

11 Up / Down Genes

SCITRAM

Mean LFC across all clusters

gene	cluster 1	cluster 2	cluster 3	cluster 4	cluster 5	cluster 6	cluster 7	cluster 8	cluster 9	mean_LF
1 SOS1	1.30	-0.79	-1.28	-1.31	-0.62	-1.39	0.48	1.78	-2.02	-0.43
2 IFITM2	1.12	-1.10	-0.85	-1.63	0.014	-1.08	1.19	0.83	-1.07	-0.29
3 PPIB	0.99	-0.069	-0.67	-0.82	-0.63	-0.97	1.18	0.47	-1.17	-0.19
4 GZMA	0.99	-1.47	-0.64	-1.29	0.045	-0.57	1.65	0.29	-2.03	-0.34
5 CD52	0.95	-0.89	-0.034	-0.22	-0.45	-0.61	0.91	0.3	-1.12	-0.13
6 IKZF1	0.93	-0.33	-0.77	-0.51	-0.44	-0.39	0.77	0.73	-1.28	-0.14
7 CALR	0.93	0.12	-0.59	-0.71	-0.54	-0.82	1.05	0.33	-1.13	-0.15
8 EBP	0.9	-0.098	-0.52	-0.55	-0.23	-0.64	0.68	0.29	-0.59	-0.084
9 S100A6	0.89	0.014	-0.91	-0.85	0.039	-0.73	0.57	0.52	-0.54	-0.11
10 MLLT3	0.89	-0.33	-0.57	-0.41	-0.66	-0.51	0.53	1.10	-1.57	-0.17
11 CTSC	0.88	-0.15	-0.65	-0.056	-0.21	-0.4	0.66	0.1	-0.89	-0.079
12 BCL2	0.88	-0.63	-0.1	-0.0057	-0.54	-0.083	0.48	0.56	-1.83	-0.14
13 THEMIS	0.87	-0.62	-0.42	-0.12	0.00022	-0.041	0.022	0.83	-1.73	-0.13
14 HSPA5	0.87	-0.17	-0.75	-0.83	-0.74	-0.76	1.27	0.45	-0.9	-0.17
15 HSPA8	0.85	-0.22	-1.18	-1.22	-0.96	-0.024	1.17	0.71	-1.04	-0.21
16 DPP4	0.83	-0.28	-0.89	-0.97	-0.15	-0.7	0.82	0.84	-0.81	-0.15
17 HTATSF1	0.82	-0.33	-0.53	-0.51	-0.41	-0.6	0.91	0.42	-0.68	-0.1
18 PLXNA4	0.82	-0.49	-0.4	-0.32	0.031	-0.37	0.52	0.18	-0.46	-0.056

Export table

11 Up / Down Genes

SCITRAM

Cluster 2

gene	lfc
1 AHI1	1.81
2 RBPJ	1.57
3 LYST	1.47
4 APBB2	1.42
5 PAM	1.41
6 MT2A	1.34
7 DUSP4	1.32
8 FLNA	1.31
9 TNFRSF9	1.29
10 ATP8B4	1.28
11 AL138720.1	1.28
12 PDE3B	1.26
13 MTSS1	1.25
14 TIAM2	1.24
15 FABP5	1.22
16 IFNG	1.21
17 CDK6	1.20
18 PFN1	1.20

Export table

SCITRAM has been designed for reconstructing gene regulatory networks from differential gene expression readouts assessed between selected clusters. Hence, when clicking Network Options (14) on the main window, you can set up the network analysis keeping values by default or changing it, keep in mind that the LFC provided by default is only a suggestion, all other values could be keep it in first instance.

Clusters:

1	Dataset summary
2	Total cells: 27217 (100%)
3	Select a cluster to view detailed information.
4	
5	
6	
7	
8	
9	
10	
all	

Search Gene:

Differential gene expression analysis:

14 Network setup

1) Select Cell Clusters:

Reference cluster: Target cluster:

1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10

2) or

Down-regulated < -0.2 <= non-responsive <= 0.2 < Up-regulated

☐ Keep no responsive genes

3) Select Gene Regulatory Network Collection:

CellNet_Human_GRN032014_PTOX

4) Set Edge Correlation

corr > 0

5) Confidence by GRN randomization:

Number of Iterations 20 P-value < 0.05

a) A panel allows to select the target and the reference cluster for the differentiation expression analysis.

b) Define the fold-change (in log2 scale) range threshold to be used for defining differentially expressed genes. The choice of the LFC is important because it is going to define the number of genes that are going to participate at the reconstruction of the gene regulatory maps. We use 0.2 for this example (0.5 default)

c) If you are not sure about which LFC range you need to use, you can test several LFCs from a range and steps, this will be explained below.

2) or

Calculate LFC : From 0.1 to 1 by step of 0.1

d) Selecting this option, you can use genes with lower LFC than the range selected to calculate the MRI (yield value).

e) Select a regulon collection, composed by TF-TG relationships collected from the public domain. Currently we provide a set from CellNet ([Cahan et al; Cell 2014](#)) to analyze Human or Mouse samples. If you want to add another collection, simply add a file in the “GRN collection” directory, the columns are “Target Gene”, “Transcription Factor” and some “score”.

f) Filter for Edges: This value depends on the GRN collection. In CellNet it corresponds to the correlation parameter “corr” and allows to filter the edges with lower correlation values than corresponding threshold (symmetric threshold)

g) This parameter allows to randomize the TF-TG connections to evaluate whether the network connectivity is relevant for the system. You can select the number of randomization events (iterations) and the accepted P-value.

h) Run the process could take several minutes depending of the LFC selected and the number o iterations, a new window will appear finishing the analysis, the main network window.

To help with the choice of the LFC, SCITRAM allows to explore the consequence of various LFC thresholds. When clicking on “Test several LFCs” **(c)**. Then a table containing the LFC threshold in use is provided, the number of differentially expressed genes (DEG), number of Nodes (genes in a network), number of edges (connections between genes), the Density where a higher value corresponds a more edges connected to the nodes, and number of components, in 1 component you have only one main network, more than one you start to have satellite networks. The row In blue corresponds to a network with higher density and one point before to have satellite networks. This blue arrow is a suggestion, and you need to test if this LFC shape with your analysis. Like in **(h)** “Run Network with selected LFC” will show the next main network window.

c+h
Test several LFC window

autoLFC – Network exploration

Networks by LFC

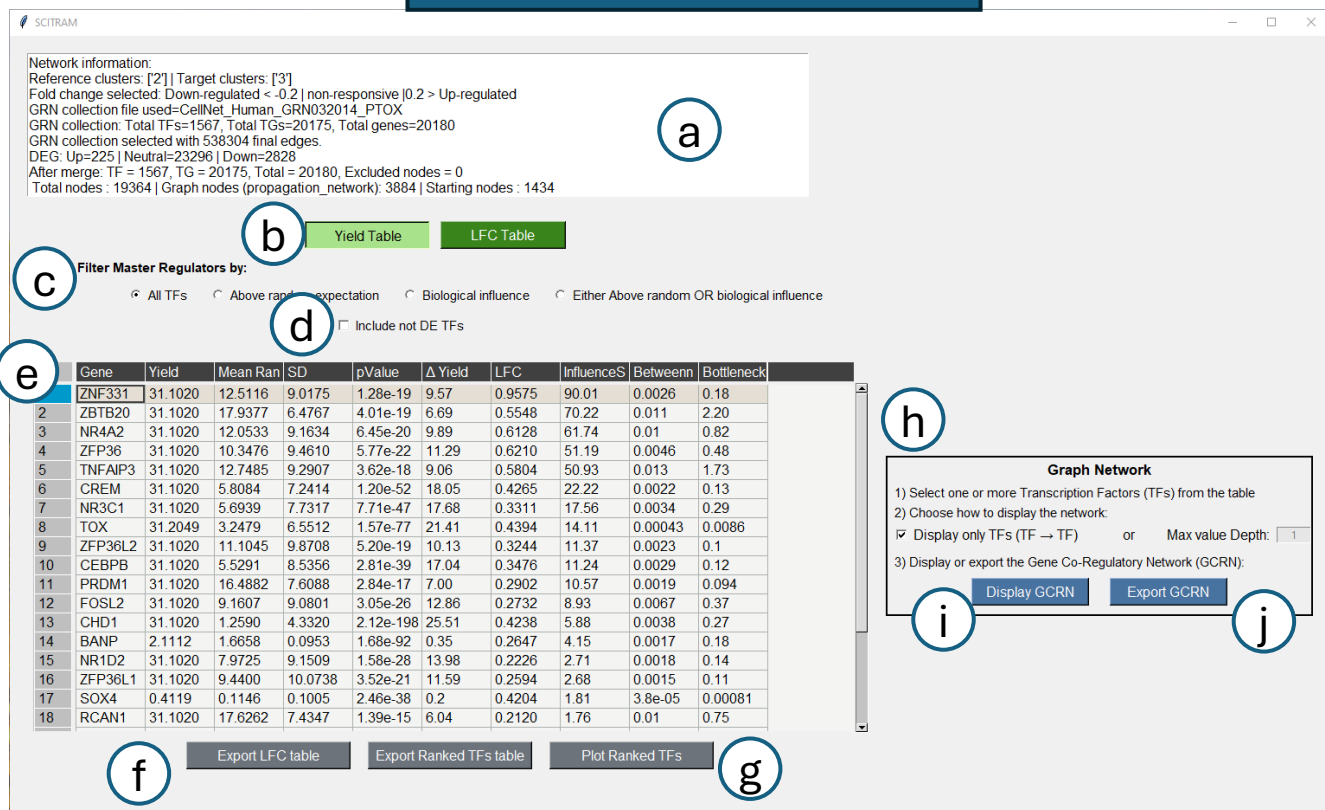
	LFC	DEG	Nodes	Edges	Density	Component
1	0.1	5563	5968	45926	0.0013	1
2	0.2	3032	3877	25841	0.0017	1
3	0.3	1714	2743	15048	0.002	2
4	0.4	992	2074	9150	0.0021	4
5	0.5	568	1546	5440	0.0023	6
6	0.6	350	1213	3581	0.0024	5
7	0.7	210	900	2162	0.0027	4
8	0.8	117	640	1230	0.003	3
9	0.9	72	509	895	0.0035	3
10	1.00	45	384	559	0.0038	3

Blue row: Recommended LFC based on network connectivity stability

Run Network with Selected LFC

The main network window shows a Transcription factors (TFs) ranked in the number of target genes under their control over the whole reconstructed Gene regulatory network (GRN) called “yield”.

Main Network window



Network information:
Reference clusters: [2] | Target clusters: [3]
Fold change selected: Down-regulated < -0.2 | non-responsive | 0.2 > Up-regulated
GRN collection file used: CellNet_Human_GRN032014_PTOX
GRN collection: Total TFs=1567, Total TGs=20175, Total genes=20180
GRN collection selected with 538304 final edges.
DEG: Up=225 | Neutral=23296 | Down=2828
After merge: TF = 1567, TG = 20175, Total = 20180, Excluded nodes = 0
Total nodes : 19364 | Graph nodes (propagation_network): 3884 | Starting nodes : 1434

Filter Master Regulators by:
☒ All TFs ☐ Above random expectation ☐ Biological influence ☐ Either Above random OR biological influence
☐ Include not DE TFs

Gene	Yield	Mean Rank	SD	pValue	Δ Yield	LFC	InfluenceS	Betweenness	Bottleneck	
1	ZNF331	31.1020	12.5116	9.0175	1.28e-19	9.57	0.9575	90.01	0.0026	0.18
2	ZBTB20	31.1020	17.9377	6.4767	4.01e-19	6.69	0.5548	70.22	0.011	2.20
3	NR4A2	31.1020	12.0533	9.1634	6.45e-20	9.89	0.6128	61.74	0.01	0.82
4	ZFP36	31.1020	10.3476	9.4610	5.77e-22	11.29	0.6210	51.19	0.0046	0.48
5	TNFAIP3	31.1020	12.7485	9.2907	3.62e-18	9.06	0.5804	50.93	0.013	1.73
6	CREM	31.1020	5.8084	7.2414	1.20e-52	18.05	0.4265	22.22	0.0022	0.13
7	NR3C1	31.1020	5.6939	7.7317	7.71e-47	17.68	0.3311	17.56	0.0034	0.29
8	TOX	31.2049	3.2479	6.5512	1.57e-77	21.41	0.4394	14.11	0.00043	0.0086
9	ZFP36L2	31.1020	11.1045	9.8708	5.20e-19	10.13	0.3244	11.37	0.0023	0.1
10	CEBPB	31.1020	5.5291	8.5356	2.81e-39	17.04	0.3476	11.24	0.0029	0.12
11	PRDM1	31.1020	16.4882	7.6088	2.84e-17	7.00	0.2902	10.57	0.0019	0.094
12	FOSL2	31.1020	9.1607	9.0801	3.05e-26	12.86	0.2732	8.93	0.0067	0.37
13	CHD1	31.1020	1.2590	4.3320	2.12e-198	25.51	0.4238	5.88	0.0038	0.27
14	BANP	2.1112	1.6658	0.0953	1.68e-92	0.35	0.2647	4.15	0.0017	0.18
15	NR1D2	31.1020	7.9725	9.1509	1.58e-28	13.98	0.2226	2.71	0.0018	0.14
16	ZFP36L1	31.1020	9.4400	10.0738	3.52e-21	11.59	0.2594	2.68	0.0015	0.11
17	SOX4	0.4119	0.1146	0.1005	2.46e-38	0.2	0.4204	1.81	3.8e-05	0.00081
18	RCAN1	31.1020	17.6262	7.4347	1.39e-15	6.04	0.2120	1.76	0.01	0.75

Graph Network
1) Select one or more Transcription Factors (TFs) from the table
2) Choose how to display the network:
☒ Display only TFs (TF → TF) or Max value Depth: 1
3) Display or export the Gene Co-Regulatory Network (GCRN):

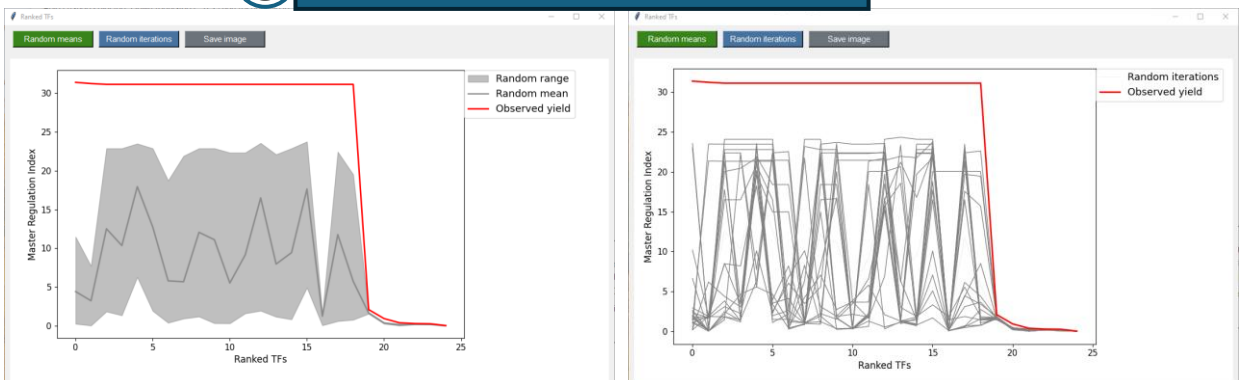
Export LFC table **Export Ranked TFs table** **Plot Ranked TFs**

- Information section shows information about the values used for the analysis, clusters, LFC, etc, and specifics for the results like number of nodes, target genes, and starting nodes, you can copy paste this info.
- There are two options to show in the main table **(f)** the Yield by default and the LFC table optional, for the last one, all the options in **(c, d, f, g, h)** are blocked.
- Filter Transcription Factors. You will see the main table with all TFs even if the “control” randomization yield is higher than the test. You can select the “Above random expectation” to show only the higher yields than the randomized yield (positive Delta Yield). You can show the TFs with “biological influence” based in the influence score (explained later) and ranked by this. Finally, you can select “Either random OR biological influence” to maximize the TFs enriched by these two values.
- Include no DE TFs shows all the TFs even if this TFs aren’t differentially expressed respect to the LFC used for the analysis, that’s is, you will see TFs with LFC lower than the threshold, this could help to see TFs low expressed but maybe with high yield.

- e) Main Table, this is an interactive table, you can copy paste values, select rows or columns and search values (with ctrl+f) here you can see the next values:
- Gene: In this table this corresponds to the transcription factor (TF).
- Yield: Percentage of genes under the control of the associated TF (direct or Indirect connections).
- Mean Random Yield: average of yields obtained over multiple randomized GRNs.
- SD: Standard deviation associated to the yields assessed over multiple randomized GRNs.
- P-value: Confidence assessed based on the multiple randomized GRNs (z-test).
- Delta yield: the difference between the Yield tested and the mean random.
- LFC: Differential expression (Log2) corresponding to the TF.
- Influence score: A composite metric quantifying a node's regulatory impact by integrating its multi-depth propagation reach (yield_d1/d2/d3) with LFC-weighted influence.
- Betweenness: A centrality measure indicating how often a node lies on shortest paths between other nodes, reflecting its role as a mediator in information flow.
- Bottleneck: A topology score combining betweenness and node degree to highlight highly connected nodes that also act as critical bridges in the network.
- f) Export the yield or LFC table, the export table will be the same as you see in the window, using the same filters and rankings.
- g) Plot Ranked TFs. The TFs with delta yield positive will be selected and ranked by yield, the mean and standard deviation range will be shown. You can see in this window 3 buttons, “random means” by default, “random iterations” to see each iteration and you can save this figure.

g

Plot ranked TFs supp window



- 

Network controls

Font size

Node size (scale)

Edge width

Zoom

Layout

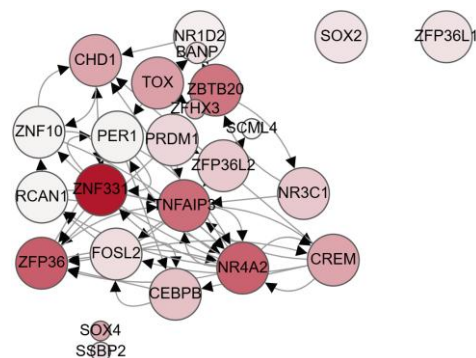
Repulsion

Node spacing

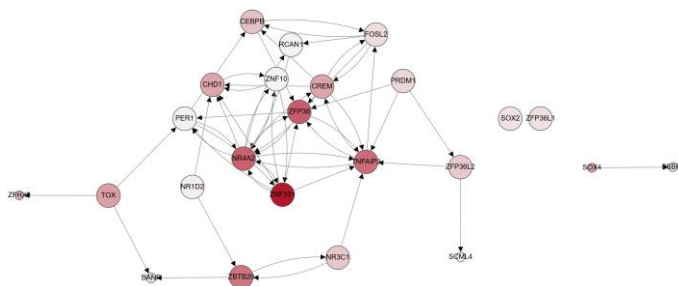
The nodes could be manipulated in the space with the mouse and doing one click, more information will be displayed (yield and LFC values). Node colors corresponds to the LFC from white to red (min to max), and the size for the yield (min to max).

Layouts:

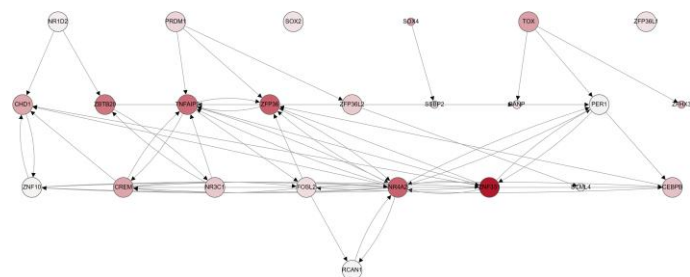
Cose:



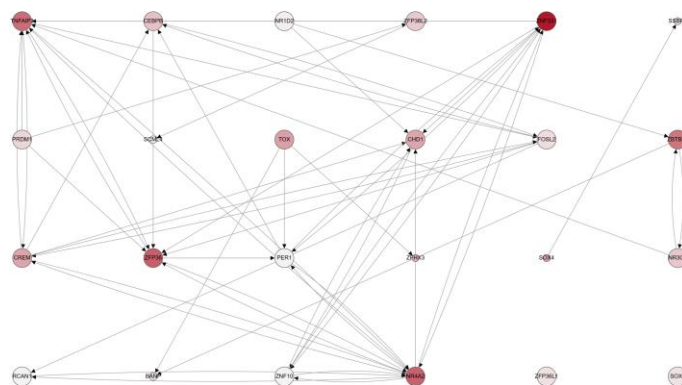
Fcose:



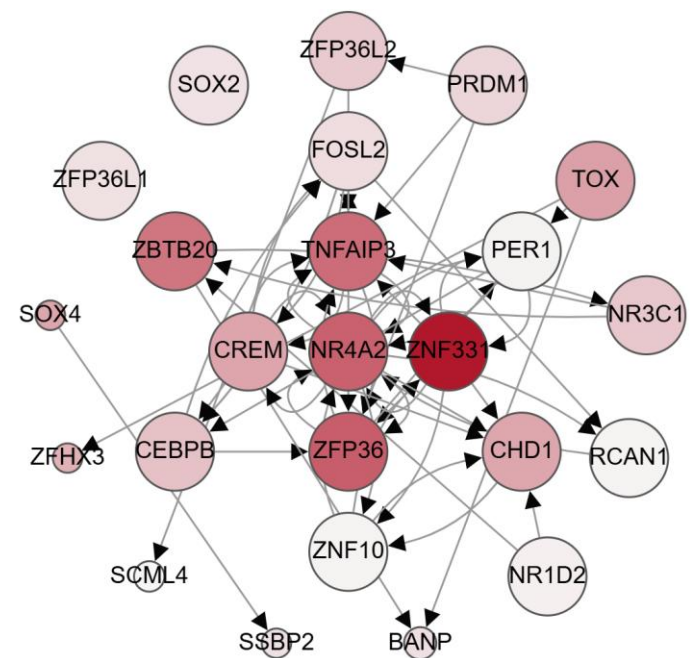
Breaththfirst:



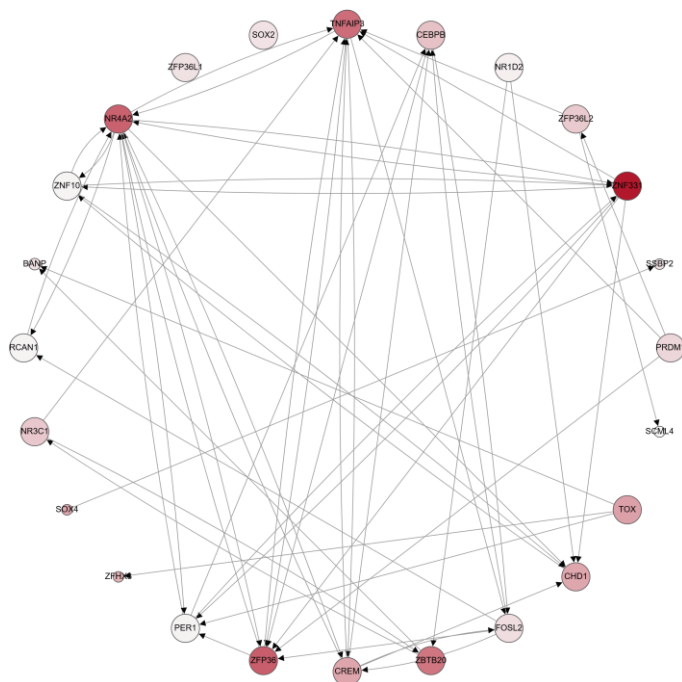
Grid:



concentric:



circle:



Network selecting TFs with delta yield positive and
Max depth =1:

